

JACOB BARNETT

Full Stack Engineer

CONTACT

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FRONTEND

React
React Native / Expo
JavaScript ES6+
HTML5 / CSS3
Three.js / WebGL
Redux / Tailwind

BACKEND

Node.js / Express
REST APIs
Firebase / Firestore
SQL / Knex.js
Auth / JWT
Serverless Fns

EMERGING TECH

Rust
WebAssembly (WASM)
NASA APIs
Physics Simulation
Real-Time Data

TOOLS

Git / GitHub
Vercel
VS Code
Agile / Scrum

EDUCATION

Full Stack Certification
BloomTech, 2024

Jacob Barnett

Full Stack Software Engineer

SUMMARY

Full Stack Software Engineer with a BloomTech Full Stack Certification and a strong record of independently designing, building, and deploying complete applications. Specializes in React, REST APIs, and data driven dashboards. Recent work includes BizWatch Connect, a live business alert and messaging app, a Rust and WebAssembly asteroid mining simulator, and a live NASA API orbital visualization, all deployed and live at jacobbarnett.dev. Brings an operations and client facing background that adds strong communication, reliability, and ownership of results.

PROJECTS

BizWatch Connect

[bizwatch.app](#) | [React](#), [React Native / Expo](#), [Firebase](#), [Firestore](#)

Business focused alert and messaging platform connecting local businesses for real time communication, safety alerts, and deals, built as a web app and a companion React Native mobile app
Discord style channels for group messaging alongside direct one to one messaging between business owners
Interactive business discovery map showing nearby businesses on the network, with the ability to message directly from the map or search by name
Firebase Authentication and Firestore backend with account deletion, data privacy controls, and rate limited alert posting to prevent spam

Asteroid Mining Simulator

[jacobbarnett.dev/mining](#) | [React](#), [Rust](#), [WebAssembly](#), [Three.js](#), [NASA API](#)

Rust physics engine implementing the Tsiolkovsky rocket equation and Hohmann transfer delta V, compiled to WebAssembly and executed entirely in browser with no backend
Six step manual mission control: Transfer Burn, Approach and Rendezvous, Anchor Deployment, Drilling, Ore Extraction, and Return Burn, each with simulated signal comm delay based on real asteroid distance
Procedural 3D asteroid generation across five shape presets, including Eros, Vesta, Itokawa, a contact binary, and Bennu style, with moon surface texture and sine wave vertex displacement
Live telemetry panel tracking fuel burn rate, ore accumulation, power and thermal fluctuations, attitude control status, and mission scoring, all driven by the Rust WASM state machine

Asteroid Mining Dashboard (AMD)

[jacobbarnett.dev/asteroids](#) | [React](#), [Three.js](#), [NASA JPL Small Body API](#), [Vercel Serverless](#)

Fetches sixty real near Earth asteroids from NASA's JPL Close Approach API through a custom Vercel serverless proxy built to bypass CORS restrictions
Interactive 3D Earth, Moon, and asteroid orbital visualization with click to zoom inspection, color coded PGM potential, and adjustable delta V and diameter filters
Selected asteroids pass directly into the Mining Simulator through React Router state, creating a seamless two app mission planning workflow

Tesla Car UI Dashboard

[jacobbarnett.dev/tesla](#) | [React](#), [Spotify OAuth 2.0](#), [Three.js](#), [REST APIs](#)

Full OAuth 2.0 Spotify integration with live playback controls, track metadata, and album art, authenticated through a PKCE flow with token refresh
3D low poly Tesla Cybertruck model rendered in Three.js with animated panels, plus climate, navigation, and vehicle settings panels with smooth state transitions

RoboTaxi Fleet Monitor

[jacobbarnett.dev/robotaxi](#) | [React](#), [Leaflet.js](#), [Real Time Simulation](#)

Live operations dashboard simulating a fleet of twenty autonomous Tesla Cybercabs across Los Angeles on an interactive Leaflet map
Real time vehicle tracking, passenger pickup and dropoff simulation, battery level monitoring, and dispatch status, all updated on a live interval loop

EXPERIENCE

Roofing Installer and Manager

[Barnett Roofing](#), Yorba Linda, CA | [March 2022 to December 2025](#)

Managed client communications, project timelines, and team scheduling for residential and commercial jobs while self studying software engineering outside of work hours

Assistant General Manager

[Crunch Fitness](#), Huntington Beach, CA | [July 2018 to January 2021](#)

Led a sales team to exceed quarterly targets through coaching, data tracking, and strategic outreach campaigns